

Business Case

Perfect Purifiers LLC | Product Development Planning | Written by Brian McCarthy

Business Case

Section	Description
Market Situation	The air purifier market is growing and customer expectations are rising. HEPA purifiers remain popular, but many customers are dissatisfied with current product limitations such as noise, ozone emissions, odor limitations, expensive filter replacement, maintenance needs, and limited coverage. PP LLC has more than ten years in air purification and already sells HEPA purifiers, spray air purifiers, and commercial air scrubbers. This gives the company a credible base from which to improve its HEPA line.
Opportunity	PP LLC can benefit by launching an improved HEPA product line before the summer hazardous-air season. The opportunity is to win customers who want safe, efficient, reasonably priced, lower-maintenance purification across multiple price tiers. Marketing assumes the new product line can increase revenue by 25%, expand the existing market by 10%, and reduce product returns by more than 50%.
Envisioned Solution	The envisioned solution is a three-model HEPA purifier family: Basic, Better, and Ultimate. The solution combines reduced noise, reduced or eliminated ozone, increased motor strength for wider coverage, a longer-life lower-cost filter, and an air freshener/cleaning option. The team arrived at this solution by combining customer survey pain points, engineering feasibility, operations channel capability, and sponsor expectations for return on investment. This is better than alternatives because it addresses multiple customer problems rather than optimizing only one feature.
Objectives and Assumptions	Objectives: increase market share, sales, revenue, and customer satisfaction; reduce returns; complete launch readiness within six months; validate three prototypes with five priority features; and prepare value propositions and positioning statements. Assumptions: customer concerns will remain strong through the summer market window, existing channels can support launch, engineering can meet the phased design/development schedule, prototype cost remains near \$80,000, and annual returns exceed \$200,000 for two years.
Costs	Initial Conceive and Plan funding is \$30,000. Prototype development is estimated at \$80,000 for three distinct prototypes with five features. Required resources include product management, marketing research, engineering, design, product ownership, project management, operations, supply chain support, beta testers, channel partners, procurement, and sponsor review. Long-term investment may include production tooling, inventory, packaging, retailer education, launch marketing, customer support, and replacement filter supply.
Strategic Alignment	The product aligns with PP LLC's vision to expand sales and market presence by offering improved air filtration technology to current and future customers. The concept leverages existing HEPA capabilities, operations knowledge, and distribution channels while extending the company toward higher customer value, stronger differentiation, and measurable business outcomes.
Timing	The current scenario date is January 5. The schedule targets design in January-February, development March-May, beta testing in June, and launch readiness within six months for the summer season. Timing maps directly to the market need because customers are increasingly concerned about hazardous air conditions, smoke, allergens, odors, and indoor air safety. Investment should proceed in controlled gates: concept approval, business case approval, prototype build, beta readiness, and launch readiness.
Cost Benefit Analysis	Initial planned investment is approximately \$110,000 when combining \$30,000 for Conceive/Plan and \$80,000 for prototypes. Expected returns exceed \$200,000 annually for two years, or at least \$400,000 gross benefit. That implies at least \$290,000 net benefit before long-term production and launch expenses, a benefit-cost ratio of approximately 3.6x, and a simple ROI of about 264% on the initial planned investment. If PP LLC does not proceed, it risks missing the summer market window, losing share to competitors, and continuing returns caused by unresolved customer pain points.
Risk	Major risks include engineering delays, cost overruns, supply chain issues, distribution channel price increases, prototype performance failure, beta feedback revealing unresolved maintenance or safety concerns, and competitor response. Mitigations include Sarah Newton maintaining a risk log, weekly team review, early supplier engagement, phased prototypes, sponsor gate reviews, alternative component sourcing, channel price monitoring, and clear go/no-go criteria before launch.
Initial Investment and Recommendation	Recommendation: approve the Product Concept and Business Case and proceed to prototype development under gated control. Initial investment requirement is \$30,000 already allocated for Conceive/Plan plus \$80,000 for prototypes. Long-term investment should be approved after beta testing confirms safety, noise, coverage, filter, and customer satisfaction targets. The business case supports proceeding because expected returns, strategic alignment, market timing, and customer problem fit outweigh the managed risks.